

Balasubramanian K

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Professional Summary

Passionate computer science undergraduate with interest in exploring applications of AI and Machine Learning Research, who has experimented in 3D scenery generation pipelines, maps navigation and related API and animation generation among other domains. Skilled in designing full stack websites and eager to deepen expertise in Computer Vision and Machine Learning through collaborative research and innovative projects.

Education

International Institute of Information Technology, Hyderabad, July 2025 – Jul 2029
B.Tech and M.S in Computer Science and Engineering (lateral entry)

Indian Institute of Information Technology Vadodara-International Campus Diu,
Daman and Diu Aug 2023 – Jul 2025
B.Tech in Computer Science and Engineering (Discontinued)

Delhi Public School, Bangalore-South May 2021 – Mar 2023
Grade XII CBSE

Prakriya Green Wisdom School, Bangalore May 2016 – Mar 2021
Grade X ICSE

Experience

Research Intern – 3D graphics generation from text input May 2025 – August 2025
National Institute of Technology Calicut
Mentor: Dr. Santosh K Behera, Department of Computer Science, NIT Calicut

- Explored stable Diffusion pipeline for generation of 2D images and Monocular depth estimation to predict depth values of generated image pixels
- Using Depth image based rendering (DIBR) to create alternate angled views of generated image
- Used image diffusion based inpainting model assisted by OpenAI's CLIP model to inpaint surrounding regions of alternate images which had no pixel information
- Invoked NeRF (neural radial fields) to train and optimize both CLIP and depth losses to inpaint the 2D images and combine them to give 3D perspective
- Implemented the pipeline on free-to-use cloud GPUs (Google Colab) by reducing resolution and batch size and by using checkpointing to stay within usage limits

Projects

Two-Stage Automatic License Plate Recognition for Indian Multi-State Vehicles
Python, YOLOv8, PaddleOCR, OpenCV, CLAHE, Streamlit, Hugging Face Spaces, PyTorch

- Built an end-to-end ALPR pipeline combining YOLOv8n object detection with PaddleOCR text recognition, targeting Indian license plates across all 35 states and union territories
- Trained YOLOv8n on a merged multi-state dataset of 9,568 annotated images sourced from Roboflow Universe, achieving 75.12% exact-match accuracy on a 414-image evaluation set – a substantial improvement over the 58% Tesseract baseline
- Designed a deterministic preprocessing chain (grayscale conversion, bicubic 2x upscaling, CLAHE contrast enhancement, Gaussian blur) to normalize plate crops for improved OCR input quality
- Implemented a multi-stage post-processing pipeline: sanitization, positional format masking with a visually-motivated confusable-correction dictionary, and a visual Levenshtein distance algorithm with character-group costs for OCR error correction
- Added semantic RTO validation using a nested dictionary of all valid Indian state–RTO district combinations, with auto-correction and flagging of invalid outputs for manual review
- Deployed the complete system as a Streamlit web application on Hugging Face Spaces, supporting single-image, batch, and video inference modes with CSV export; all training code released publicly on GitHub

Tourism Website for Diu

React, Node.js, Express.js, MySQL, SMTP, OpenStreetMap API, Software Engineering

- Built an end-to-end tourism web platform for tourists visiting Diu.
- Developed a location-based forum with dedicated pages for each tourist spot, including detailed guides and a user rating and comments system.
- Implemented a drag-and-drop personalized tour planner with email-based reminders using SMTP.
- Integrated geolocation-based services to find the nearest dustbins and restrooms, with navigation routes using OpenStreetMap APIs.
- Designed an admin dashboard to manage tourist locations, hotels/restaurants/transport options, moderate user comments, and broadcast emergency notifications via email.

DOCS++ - a text editor cum distributed file sharing system

C, Socket Programming, Networking, TCP/IP and UDP, pthreads

- Designed a 3-server architecture – Name server, Client server and Storage server using TCP socket programming in C
- Implemented metadata management using Name Server to handle file registry, user routing, and request coordination across multiple storage servers
- Built Storage Server modules supporting core CRUD operations with AVL-tree based indexing for efficient lookup
- Added sentence-level locking for safe concurrent access and multi-user file editing.
- Implemented fault tolerance using a heartbeat mechanism and replication (RAID1-style) and automated recovery sync for crashed/rejoined servers
- Integrated access control (read/write permissions) and features like checkpointing/version rollback and batch execution

Relevant Coursework

- **Statistical Methods in AI:** Data pre-processing, PCA and dimensionality reduction, GMMs, SVM, Neural Networks, CNNs, RNN & LSTM, Transformers

- **Computer Graphics:** Graphics pipeline, Transformations, Model-View-Camera-projection-clipping algorithms, OpenGL implementation, Rasterization, Ray Tracing, Lighting and shading, Bounding Volume Hierarchy and other accelerating structures
- **Machine Learning:** Clustering and K-means, Linear regression, Bayesian models, Decision Theory and Maximum Expected Utilities, Multi-Arm Bandits, A* search algorithm, Decision Trees
- **Introduction to Programming:** Basics of programming in Python; Data structures such as Lists, Tuples, Dictionaries; Object-Oriented Programming concepts
- **Database Management Systems:** ER diagrams, Normalization, SQL programming, Concurrency and deadlocks, B and B+ trees, efficient storage practices, fault tolerance
- **Data Structures & Algorithms:** Object-Oriented Programming, Trees, Heaps, Sorting, Hashing, Graphs, Dynamic Programming
- **Probability and Statistics:** Sample space, Sigma algebra, Bayes theorem, Random Variables, Expectation, Variance, Random vectors, Markov chains, MLE, Bayesian statistics
- **Discrete Mathematics:** Logic, set theory, graph theory, combinatorics, discrete probability
- **Linear Algebra:** Transformation of vectors, Eigen value decomposition, rank of matrices, vector space, SVD

Technical Skills

Programming Languages:	Python, SQL, C, JavaScript, C#, C++
Databases:	MySQL, MongoDB
Frameworks:	PyTorch, Scikit-learn, Hugging Face, NumPy, Tkinter, React JS, Node JS, Express JS, Matplotlib, OpenGL
Tools and Platforms:	VSCode, Jupyter Notebook, Linux Shell, Unity Engine, Arduino

Positions of Responsibility

- **Joint Secretary, Debate Club** Aug 2024 – April 2025
IIIT Vadodara - International Campus Diu
 - Led a team of 8 members to organize debate workshops and activities.
 - Organized institute cultural fest Debate Competition and Model United Nations event with 50+ participants.
- **Core Member, Coding Club** Aug 2024 – April 2025
IIIT Vadodara - International Campus Diu
 - Delivered a Data Structures workshop for the fresher batch (102 students).